

Modular Integrated Construction

Introduction

Modular Integrated Construction (MiC) is a construction method that employs the technique of having freestanding volumetric modules (with finishes, fixtures, fittings, etc.) manufactured off-site and then transported to site for assembly. Proven benefits include improved site safety, more efficient and better quality control, shortened construction period, less construction waste, less demand for on-site labour, less disturbance and nuisance to the neighbourhood, etc., not just contributing to the quality and sustainable built-environment but also help ease some of the challenges of the local construction industry. To encourage MiC, the Buildings Department (BD) has formulated streamlined measures and guidelines to facilitate the industry in meeting the relevant standards and requirements under the Buildings Ordinance (BO).

Considerations Unique to MiC

2. Similar to the use of prefabricated building components, the project team should engage the MiC suppliers at the early design stage to sort out the issues usually not encountered in conventional in-situ construction. Apart from the extent of standardisation and buildability of such modules, the mode of delivery with due regard to the specific site conditions, the issues that may arise from meeting the relevant requirements including those on supervision as well as the programme of plan submissions to BD should be considered in advance. General guidelines on the design and quality control requirements under the BO for MiC are given in Appendices A and B respectively.

Pre-submission Enquiry

3. Authorized Persons (AP) and Registered Structural Engineers (RSE) are encouraged to make use of the established mechanism of pre-submission enquiry service mentioned in PNAP ADM-19 to clear with BD in the early design stage unconventional design or performance of a modular prototype for acceptance under the BO before preparing the detailed designs. A determination would be available within 45 days.

Pre-acceptance of MiC Systems/Components or Prototypes

4. Furthermore, BD has set up a pre-acceptance mechanism for granting in-principle acceptance (IPA) to individual MiC systems/components or prototypes on specific performance as listed in Appendix C. On the condition that the pre-acceptance application is accompanied with the required information and the MiC systems/components or prototypes are acceptable in meeting the standards and requirements under the BO and its subsidiary regulations, BD will advise its determination within 45 days from the date of receipt of the application by issuing a letter of IPA to AP/RSE. Otherwise, BD will issue a letter of comments on the application to AP/RSE also within 45 days.

/AP/RSE

AP/RSE should clarify/respond to all comments in 30 days from the date of the letter of comments, or the application will be considered as abandoned. Within 30 days of receipt of the responses/supplementary information from AP/RSE, BD may accept the application by issuing a letter of IPA, or if the responses/supplementary information cannot satisfactorily address BD's comments, BD will refuse¹ the application with a list of outstanding items/comments.

5. To facilitate the preparation of MiC systems/components or prototypes, applicants are encouraged to discuss with BD before submitting an application. Details of the pre-acceptance mechanism are available in BD's website www.bd.gov.hk. Nevertheless, the pre-acceptance is not a pre-requisite for the approval of general building plans.

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Building Authority

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¹ If the application is refused, any subsequent revised application will be processed according to the time frame as if it is a new application, i.e. BD will advise its determination within 45 days.

Design Requirements for Modular Integrated Construction

Fire Safety

The fire protection or performance of elements of construction of MiC should be addressed. Non-code-compliant designs should be justified for having equivalent performance as the prescriptive standards and, where necessary, supported by fire engineering assessments as stated in the Code of Practice for Fire Safety in Buildings.

Joints and Gaps

2. Modular constructions typically entail more joints and gaps, including those in drainage pipes and building envelope, which are prone to water seepage. Suitable design and construction should be provided to prevent ingress/penetration of water.

Structural Design

3. The requirements on the design and construction of reinforced concrete, precast concrete and structural steel given in the Code of Practice for Structural Use of Concrete, Code of Practice for Precast Concrete Construction and Code of Practice for the Structural Use of Steel also apply to MiC elements. Particular considerations should be given to the following design aspects:

(a) **Stability**

A building structure comprising modular units must possess adequate stability to resist wind and other lateral loads. Consideration should also be given to temporary stability during assembly and installation of the modular units. Adequate support should be provided at all stages of installation to ensure that stability is maintained;

(b) **Robustness and integrity**

The overall structural system comprising modular units should be designed to be robust and be able to resist disproportionate collapse. Structural integrity should be provided by tying all modular units and in-situ elements together horizontally and vertically. The modular units should be designed to be capable of sustaining accidental loading in such a way as to prevent the extent of any resulting collapse of individual modular unit being disproportionate to the cause of the collapse;

/(c) ...

(c) Design for temporary stages

Structural action and load path may be different during the temporary stages of construction, including fabrication, lifting, transportation, assembly and installation, and may result in higher stresses in individual MiC elements. Consideration should be given to the temporary imposed loading on the modular units during each stage of construction. The temporary stability and integrity of the modular units at all stages of construction should be ensured; and

(d) Design for movements

The MiC system should be designed to accommodate cumulative movements, usually by the provision of joints, occurred throughout the various construction stages and during its service life. Consideration should be given to the various causes of movements such as creep deformation, shrinkage, differential movement, thermal movement, etc. with due regard to the fact that such movements may be short-term or long-term and may or may not be coexistent or occur concurrently. Allowance should also be made for fabrication and installation tolerances.

Provisions for Maintenance

4. The provision of access points, inspection pits or accessible recesses to facilitate inspection and repair/replacement of structural connections, drainage pipes, building services and joints should be considered during the early design stage of the MiC system.

5. The design of a MiC module as a separate balcony or utility platform (UP) module to be assembled with in-situ joints on site will impose burden on long-term maintenance, thus should be avoided. In case where a separate balcony or UP module is proposed, the area of such balcony or UP module will not be accepted as MiC floor area in the context of Joint Practice Note No. 8 jointly issued by the Buildings Department, Lands Department and Planning Department for the purpose of gross floor area and site coverage concessions.

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Quality Control and Supervision of MiC

Quality Assurance Scheme

Modular units are to be fabricated by a factory with a valid Full Certificate issued by the Building Technology Research Institute under the Modular Integrated Construction Manufacturer Accreditation Scheme (Accredited MiC Manufacturer), ISO 9001 or equivalent quality assurance certification. This will be imposed as a condition under item 6 in section 17(1) of the Buildings Ordinance (BO) when giving approval of plans.

2. Upon approval of plans, a requirement will also be imposed under regulation 10 of the Building (Administration) Regulations (B(A)R), to require submission of a copy of valid Full Certificate of the Accredited MiC Manufacturer or the Quality Assurance Scheme (QAS) of the MiC supplier at least 14 days before the commencement of production work in the prefabrication factory unless such has been covered in BD's in-principle acceptance and remains unchanged¹. For an application choosing to submit with a QAS, the project AP and RSE should provide a written confirmation that the QAS has adequate provisions in ensuring the quality of production complying with the provisions of the BO and the approved plans.

3. The QAS should cover the following items:

- (a) Quality control tests (including raw materials delivered to the prefabrication factory) with specification of testing personnel/body, testing standard, testing procedure and acceptance criteria;
- (b) Calibration of equipment for tests and inspections/audits;
- (c) Production process;
- (d) Frequency and extent of inspections by in-house staff and independent parties²; and
- (e) Frequency and extent of audits by in-house staff and independent parties.

/Qualified Supervision...

¹ The MiC supplier should submit a written confirmation if the QAS is based on the one accepted under the Pre-acceptance Mechanism for MiC Systems. If some items of such QAS have been amended, only the amendments are required to be submitted.

² "Independent parties" refer to parties not under the direct employment of the prefabrication factory (i.e. either a certification body under the Hong Kong Certification Body Accreditation Scheme (HKCAS) of the Hong Kong Accreditation Service (HKAS) or an inspection body under HKAS's Hong Kong Inspection Body Accreditation Scheme (HKIAS)).

Qualified Supervision

4. Approval of plans will be imposed with conditions:
- (a) Under item 6 in section 17(1) of the BO to specify that qualified supervision is to be provided by AP, RSE and Registered Contractor (RC) for the fabrication, assembly, installation and examination of modular units and pre-installed finishes; and
 - (b) Under regulation 10 of the B(A)R to require submission of a MiC Supervision Plan³ at least 14 days before the commencement of the production work in the prefabrication factory.
5. AP and RSE should assign their respective Quality Control Supervisory Team (QCST) to supervise the modular unit production work in respect of fire resisting constructions (such as fire resisting doors and fire resisting pipe collars), drainage works, structures, etc. in accordance with the requirements stipulated in PNAP APP-158. Similarly, RC should assign a Quality Control Co-ordination Team (QCCT) to supervise the modular unit production work in the prefabrication factory. AP, RSE and Authorized Signatory (AS) of RC should respectively determine the required qualification of the supervisory personnel and necessary frequency of supervision of QCST and QCCT subject to the minimum standards in Table 1 below.

Table 1 Minimum Qualification and Supervision Frequency of QCST and QCCT

	AP Stream	RSE Stream	RC Stream	
Qualifications of Supervisory Personnel	T3*	T3*	T3*	T1*
Supervision Frequency	Weekly	Weekly	Weekly	Continuous

* T3/T1 refers to Grade T3/T1 Technically Competent Person equivalent as stipulated in the Code of Practice for Site Supervision.

6. The names and qualifications of the supervisory personnel assigned by AP, RSE and RC respectively should be recorded in an inspection log book. The details of production, inspection, auditing and testing of modular units should be recorded in the log book by the supervisory personnel. The log book should be kept in the prefabrication factory and a copy of it should be kept at the building site office and, when required, produced to officers of BD for inspection.

/Quality Audit...

³ The MiC Supervision Plan should contain the names, qualifications, identification, inspection frequency, confirmation of appointment and contact information of the supervisory personnel assigned by AP, RSE and RC under paragraph 5.

Quality Audit

7. AP, RSE and AS of RC should visit the prefabrication factory in person to inspect the production of the first batch of modular units⁴ (the first visit) and subsequently carry out quality audit checks to the prefabrication factory at least once every month. The quality audit checks should comprise inspections at the prefabrication factory in person at quarterly intervals while the monthly quality audit checks in between the first visit and quarterly quality audit checks (i.e. in the 2nd, 3rd, 5th, 6th months, and so on) may be carried out either by visiting the prefabrication factory in person or by videotelephony⁵. A requirement will be imposed under regulation 10 of the B(A)R, when giving approval of plans to specify that a copy of AP, RSE and AS's quality audit reports of the prefabrication factory duly endorsed by AP, RSE and AS respectively should be submitted to BD within 14 days after completion of the quality audit checks⁶ for record purpose. These quality audit reports should also cover the qualified supervisions by the respective QCST and QCCT.

8. Quality audit checks conducted by AP, RSE and/or AS using videotelephony should comply with the following requirements:

- (a) The inspection by AP, RSE and AS should be conducted with the assistance of QCST and QCCT under their respective stream in the prefabrication factory;
- (b) The level of supervision by videotelephony should not be inferior to that carried out personally at the prefabrication factory. In addition, all supervision items covered in the videos taken should be recorded contemporaneously in the audit reports of AP, RSE and AS, and should be submitted to BD together with the DVD-ROM discs as required in paragraph 8(c) below; and
- (c) AP, RSE and AS should submit the quality audit reports with non-rewritable DVD-ROM discs of the videos to BD within 14 days after completion of the quality audit checks by videotelephony. They should certify on each disc with a permanent marker signifying that they personally used videotelephony for compliance with the approval conditions.

/Alternative Arrangement...

⁴ The purpose of the first visit to the prefabrication factory is to verify that the QAS and quality assurance/quality control procedures are duly followed and the production line is functioning properly. For precast concrete works, the inspections by RSE and AS of the first batch of precast concrete production should cover items specified in clauses 3.1 to 3.14 and 4.1 of the Code of Practice for Precast Concrete Construction.

⁵ Videotelephony means two-way simultaneous communication with both audio and video in real time through telephone or computer network connection. The video should be recorded in colour with resolution of not less than 480p in a non-rewritable DVD-ROM.

⁶ Including the first visit and each subsequent quality audit check.

Alternative Arrangement for Quality Audit Checks at Prefabrication Factory

9. As an alternative to the requirement for the quality audit checks by AP/RSE at the prefabrication factory specified in paragraph 7 above, AP/RSE may consider to carry out the first visit to the prefabrication factory in person, then conduct on-site quality audit checks after the MiC elements are delivered to the building site. In such a situation, AP/RSE is required to notify the Building Authority in writing one month before the commencement of the production work in the prefabrication factory for the adoption of the alternative arrangement. AP/RSE should, in addition to the submission of quality audit report within 14 days of the first visit, submit on-site quality audit reports of the MiC elements delivered to the building site to BD within 14 days after completion of the on-site quality audit checks. The quality audit reports should cover the qualified supervision by the QCST. Minimum requirements of the on-site quality audit checks by AP/RSE are given in the Annex.

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Minimum Requirements of AP's/RSE's On-site Quality Audit Checks on MiC Elements Delivered to the Building Site

The quality audit checks on the MiC elements delivered to the building site carried out by AP/RSE as an alternative to the quality audit checks on MiC elements at the prefabrication factory should comply with the requirements specified in paragraph 2 below.

2. The rate of sampling for the quality audit checks shall be at least 1% of each type of modular units delivered to the building site. The quality audit check for each sample of the modular units should include the following items, where applicable:

Works Item		Scope
AP		
1	The setting out and interior	● Check critical dimensions including storey height, projections, balconies, utility platforms, A/C platforms, protective barriers, internal areas, exit routes, etc. ● Check conformity with general layout
2	Installation of windows and flue apertures	● Check minimum areas of glazing and openable areas of prescribed windows and windows for rooms containing soil or waste fitment and water tightness in accordance with PNAP APP-116 ● Check positions and dimensions of flue apertures ● Check provisions required under modification/exemption
3	Provision of barrier free access	● Check provisions and critical dimensions of bathrooms, toilets, ramps, corridors, lobbies, doors, handrails, signage, tactile guide paths, etc.
4	Drainage	● Check provision of sanitary fitments ● Check drainage works including materials, dimensions, water seal traps, vent or anti-siphonic pipes and any other necessary components
5	Fire resisting construction ¹	● Check materials, fire protection of structural elements and construction of fire barriers and installation of proprietary products such as FRR of fire doors, smoke seals, self-closing devices, pipe collars, etc.
6	Finishes and fittings	● Check water proofing where appropriate such as roofs, shower areas, etc. including reports of water ponding tests/spray tests for impermeable construction ● Check provision of required mechanical ventilation and artificial lighting

¹ Open-up audit check of fire resisting dry walls is required.

		● Check provision of required fire services installations such as smoke detectors and sprinkler heads for open kitchens, etc.
7	Inspection records	● Log books of the supervisory personnel of QCST assigned by AP
8	Others	● Any other items considered essential by AP for the project
RSE		
1	Steel reinforcing bars	● Opening-up of the concrete surface at three locations, each with an area of 100mm x 300mm, to check the size, pattern and layout of the steel reinforcing bars; and the concrete covers ● Measurement of the concrete cover to steel reinforcing bars by covermeter at six locations, each with six readings taken in an area of 450mm x 450mm ● Material testing reports
2	Concrete	● Concrete strength by non-destructive test[#] (e.g. rebound hammer test) at three locations ● Material testing reports
3	Structural steel	● Non-destructive testing of welded joints by means of visual inspection, magnetic particle inspection/dye penetration inspection and ultrasonic examination, where appropriate^{2#} ● Tensile strength testing^{3#} of one specimen for every 40 tonnes of each section or plates of same thickness from the same cast to be used for fabrication of modular units selected at random by the QCST in the prefabrication factory ● Material testing reports
4	Finished products	● Concrete surface and defects ● Steel surface/corrosion protection ● Sizes, dimensions and fabrication tolerances ● Starter steel reinforcing bars ● Shear connectors
5	Inspection records	● Log books of the supervisory personnel of QCST assigned by RSE
6	Others	● Any other items considered essential by RSE for the project

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² Refer to the requirements specified in section 14.3.6 of the Code of Practice for the Structural Use of Steel.

³ Refer to the requirements specified in BS EN 10002-1:2001 or BS EN ISO 6892-1:2009.

[#] Test results should be submitted to BD within 60 days of the delivery of the modular units to the building site.

Pre-acceptance Application Checklist for MiC

This application checklist aims to remind AP and RSE of the essential information which should be contained in the plans and supporting documents accompanied with the application. The checklist should be completed by ticking the items relevant to the application. Sample drawings showing acceptable standards for the pre-acceptance application are attached at Annex I for reference.

BD will conduct regular review on this application checklist in the light of experience gained in processing different MiC systems, feedbacks from the building industry on the use of various MiC systems and technological development in the relevant fields.

- Essential information to be provided on plans
- △ Essential information to be provided in the supporting document accompanied with the plans

Section		Essential Information
1.	General	○ General notes on compliance with applicable regulations/codes of practice/design manual/guidelines. ○ Schedule/notes on intended building height, number of storeys and use of building and modules. ○ General building plans¹ (plans of all MiC floors or MiC individual system/component or prototype) in scale not less than 1:100 with full dimensions. ○ Diagrams to indicate the extent and configuration of all types² of MiC modules. ○ Structural plans¹ (plans of floors constructed by MiC, and sections/elevations) in scale not less than 1:100 showing the layout and dimensions of all structural elements at floor and ceiling levels, modular units, structural connections and locations of movement joints. ○ Plans, sections and elevations of each module type and their interfaces with upper, lower and adjacent modules/in-situ construction in scale not less than 1:50 with full dimension indicating features, if applicable, like gas flue aperture, balcony, utility platform, A/C platform, curtain/window wall, non-structural external wall system/cladding, sunken slab, impermeable construction of rooms with water supply,

¹ The pre-acceptance mechanism is applicable to individual MiC system/component or prototype. In this regard, such application accompanied with essential information should include the module(s) of the proposed MiC system/component or prototype that represents the entire building and building height when assembled.

² Different structural/building layouts and/or different dimensions of the modules constitute different types of MiC modules (e.g. mirrored modules are considered as one type of module).

Section	Essential Information
	protective barrier, projections, vertical greening, pipe ducts, etc. ○ Standard details to show the connections and methods of sealing up gaps at roof, floor, partition or internal wall, between modules and in-situ construction indicating the cavity barriers and waterproofing provisions. ○ Schedule of Finishes, Fixtures and Fittings Prefabricated/Installed Off-site. A sample schedule is attached at Annex II for reference.
2. Fire Safety	○ Demonstration on compliance with the Code of Practice for Fire Safety in Buildings (FS Code) in terms of means of escape, fire resisting construction and means of access for firefighting and rescue for MiC portion only, for example: (a) Tables to demonstrate the following: (i) Occupant capacity, required and provided no. and width of exit doors and exit routes; (ii) Construction and materials for walls, floors, columns, beams and stairs with fire resistance rating of elements of construction; and (iii) Application of fire resisting products and materials of non/limited combustibility including the references of the corresponding test/assessment reports. A sample schedule is attached at Annex III for reference; (b) Direct distance and travel distance of exit routes; (c) Fire resisting construction of fire barriers separating different occupancies, uses, compartments and the areas of special hazard; of the protected exit routes, vertical shafts and required staircases; and of the MiC portion and the interface with in-situ construction; (d) Provision of fireman's lift and firefighting and rescue stairway; and (e) Provisions in general notes on: (i) Non-combustibility requirements for cladding/ external wall/curtain wall including the insulation/water proofing materials; (ii) Limited combustibility requirements for linings/ coverings of internal wall, ceiling, floor and decorative finishes; and (iii) Non-combustibility requirements for acoustic and thermal insulation in ductings and concealed location.

Section		Essential Information
		△ Documentary proofs to illustrate compliance of FS Code, for example: (a) Test/assessment reports for the fire rated products and materials as well as combustibility of materials as necessary; and (b) Fire safety review of the system conducted by fire safety engineer, if applicable.
3.	Lighting and Ventilation	○ Table to demonstrate the provision of prescribed windows for habitable rooms and offices as well as provision of natural lighting and ventilation for rooms containing soil or waste fitments with indication of adequate window area. Detailed calculations of the provisions are not required.
4.	Drainage	○ Drainage part plans/diagrams for MiC portion only to be incorporated into the general building plans (i.e. no separate submission of drainage plans is required) to demarcate the extent of drainage works prefabricated off-site and to be carried out on site, as well as to illustrate compliance with Building (Standards of Sanitary Fitments, Plumbing, Drainage Works and Latrines) Regulations, for example: (a) Standard details and connections of traps with replenishing function for floor drains of kitchens of domestic units, toilets or pantries (PNAP APP-164); (b) Provision of surface water discharge for cantilevered structures exposed to weather; (c) No pipework for a domestic unit shall protrude into the unit under separate occupancy; (d) Common above-ground soil/waste/rainwater stacks and underground drains located in common parts of the building (PNAP APP-93); (e) No water-borne pipe embedded in structural elements (PNAP APP-105); and (f) Provision of condensate disposal system for A/C units (PNAP APP-112).
5.	Barrier Free Access	○ Provision of access and facilities for persons with a disability for MiC portion only, for example, adequate door width and provision of manoeuvring space in corridor, accessible toilets and bathrooms, if applicable.
6.	Structure	○ Structural system including design assumptions of the building and MiC, details of modular units illustrated with plans at floor and ceiling levels and sections/elevations (including recess for in-situ casting or installation works), and method of assembly and installation. ○ Material specifications and compliance standards.

Section	Essential Information
	○ Design loads (including temporary/construction loads and load effects from temperature change and top floor(s) cast in-situ/installed on-site). ○ Durability (e.g. corrosion protection) and fire protection of structural elements. ○ Quality control requirements. ○ Details of structural connections between modular units and between modular units and cast in-situ structural elements (if any). △ Vertical and lateral load transfer mechanism both within modular unit and at connections at permanent and temporary stages. △ Structural stability (e.g. resisted by both/either modular units and/or cast in-situ structural elements) at permanent and temporary stages. △ Robustness (i.e. design of vertical and horizontal ties, and hypothetical removal of critical element). △ Consideration for temporary stages (lifting, storage, installation, etc.).
7. Quality Assurance	△ Full Certificate issued by the Building Technology Research Institute under the Modular Integrated Construction Manufacturer Accreditation Scheme. - OR △ Certification on ISO 9001 quality assurance to the prefabrication factory; and △ Quality Assurance Scheme of the prefabrication factory in accordance with paragraphs 2 and 3 of Appendix B of PNAP ADV-36.
8. Fabrication & Installation	△ Description of on-site installation works with step-by-step pictorial illustrations of procedures/sequence (may be supplemented by Building Information Modelling files in digital format as specified in PNAP ADV-34) including: (a) Master construction sequence indicating cast in-situ structures and installation sequence of the MiC modules; and (b) Detailed installation process of structural connections among modular units and between modular units and cast in-situ structural elements (if any). ○ Fabrication and installation tolerances.

Section		Essential Information
9.	Maintenance	○ Accessible recesses at strategic locations for the repair/replacement/monitoring of the critical structural members and modular units connections, e.g. joint fillers, waterproofing arrangement, structural connections (e.g. bolts/welds/interlocking), etc., if any. Considerations on : (a) Access for future external maintenance and repair of external features, A/C units, cladding, etc., if any (Code of Practice on Access for External Maintenance and PNAP ADV-14); (b) Provision of pipe duct/pipe well (PNAP APP-93); and (c) Access points for inspection of drains housed in sunken slabs.

* Delete as appropriate

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